UNIT 1 · KINEMATICS · WEEK 2

Problem Set 5 — Uniform Acceleration

Physics 2026–27 · due at the start of class, Tuesday, October 6

NAME _____ DATE _____

How to work this set. Five variables — v_i, v_f, a, t, Δx — four equations, one selection rule: *find the one variable you neither know nor want, and use the equation that omits it.* Name that missing variable in your E row every time. Symbols first, numbers last, units the whole way.

1 Pick the equation — don't solve

SELECTION RULE

For each scenario, name the **missing variable** (neither known nor wanted) and write the equation you'd use. No arithmetic.

a. A jet starts from rest and accelerates at 3.5 m/s² down 900 m of runway. You want its takeoff speed.

b. A train slows from 40 m/s to 10 m/s over 60 s. You want how far it traveled.

c. A dropped phone falls for 0.80 s. You want how far it fell.

d. A sprinter reaches 9.0 m/s in 3.0 s from rest. You want her acceleration.

ANSWERS

2 The rocket sled

KDEA

A rocket sled starts from rest and accelerates at a constant $12\ m/s^2$ for $5.0\ s$. Two targets, two KDEA passes:

a. Its final velocity.

b. The distance it covers.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	WHICH EQUATION? NAME THE MISSING VARIABLE
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

3 No time given — on purpose

KDEA

A car moving at 22 m/s brakes with constant acceleration and stops in 40 m. Find the acceleration (sign included). Notice: the word "time" appears nowhere in this problem — let that steer your E row.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	WHICH EQUATION? NAME THE MISSING VARIABLE
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

4 No acceleration given

KDEA

A runner speeds up steadily from 2.0 m/s to 8.0 m/s over 3.0 s. How far does she travel during those 3.0 s? (Acceleration is constant, but its value never appears — you don't need it.)

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5 Two-stage trip

MULTI-STEP

A motorcycle starts from rest, accelerates at 4.0 m/s² for 6.0 s, then holds a constant velocity for another 8.0 s.

a. How fast is it moving at the end of the first stage?

b. How far does it travel during the first stage?

c. How far during the second stage? (Careful — constant velocity means the four-equation bank isn't even needed here.)

d. Total distance for the whole trip?

WORK

6 Speeding up or slowing down?

SIGNS

For each, state whether the object is speeding up or slowing down, using the sign rule (same signs vs. opposite signs):

a. $v = +14$ m/s, $a = -3$ m/s²

b. $v = -6$ m/s, $a = -2$ m/s²

c. $v = -9$ m/s, $a = +4$ m/s²

d. One sentence: why does "negative acceleration" alone tell you nothing about slowing down?

7 Grade this work

CRITIQUE

The problem: *"A car slows uniformly from 20 m/s to rest in 5.0 s. How far does it travel while stopping?"* A student writes:

$$\Delta x = v_i t + \tfrac{1}{2}at^2 = (0)(5.0) + \tfrac{1}{2}(-4.0)(25) = -50 \text{ m}$$

a. The equation choice is fine and a = –4.0 m/s² is correct. Find the actual error. (Hint: read the subscripts out loud, Workshop-I style.)

b. Fix it: run the full KDEA and report the real stopping distance. Sanity-check the sign of your answer.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	WHICH EQUATION? NAME THE MISSING VARIABLE
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

Graded on the homework 4–3–2–1 scale. Every E row must name the missing variable — that one habit is most of this week.